

Tanmay Sharma

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EDUCATION

Northeastern University, Boston, MA

Estimated Graduation: December, 2026

Master of Science, Cybersecurity

GPA 3.58/4.00

- Relevant Coursework: Network Security, Binary Exploitation, Reverse Engineering, CTF, Cyberlaw.

SRM Institute of Science and Technology, Chennai, India

June, 2024

Bachelor of Technology, Computer Science and Engineering

GPA 8.97/10.00

- Relevant Coursework: Operating Systems, Computer Architecture, Network Security and Offensive Security.

TECHNICAL SKILLS

- **Development:** C/C++, Python, Linux x86 Assembly(32bit/64bit), Bash, Powershell, Rust(learning), SQL
- **Tools:** Metasploit, Wireshark, GDB Debugger, Ghidra, Nikto
- **Native Hacking:** Stack & Heap Buffer Overflow, Format String Vulnerability, ROP, Ret2Libc Attack
- **Network Security:** cryptographic protocols, SSL/TLS, IPsec, OAuth, intrusion detection, vulnerability analysis, secure application development, penetration testing, cyber attack mitigation.
- **Pen Testing:** Attacks on Active Directory, OSINT, Windows and Linux Privilege Escalation
- **Misc:** Raspberry Pi, ESP32 Microcontrollers, Flipper Zero

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** NSE 1-8 certifications, Palo Alto fundamentals of network and cloud security
- **In Progress:** PNPT (Practical Network Penetration Tester), HTB CBBH Certificate
- **Achievements:** SimSpace Cyber Cup 2025 Top 6, Pwn.college white belt

WORK EXPERIENCE

Null NEU, Northeastern University

September 2024 – Present

Outreach Director, FOSS Contributor & Researcher

Boston, MA

- Contributed to the club's open-source github repository
- Organized and Outreached for Null CTF event - Hades
- Facilitated outreach to security professionals and meetups, securing guest speakers to enhance the club's events and reputation.

CyberspACe securiTy and forensIcs lab (CactiLab)

Spring 2024 – Present

Member

Boston, MA

- Working on reverse engineering the EdgeTPU driver on Pixel devices, to understand how the driver works and how it interfaces with the firmware.

Institute for Plasma Research (IPR)

September 2023 – November 2023

Project Intern

Ahmedabad, India

- Developed a secured Raspberry Pi access point functioning as a VPN/Personal Router for public Wi-Fi use.

PROJECTS

Secure Chat Application

- Secure terminal-based chat app using SRP for authentication, ECDH for key exchange, and AES-GCM for encryption, developed as part of the CY6740 course - Network Security.
- Real-time encrypted messaging with user login, online user listing, and peer-to-peer communication.
- Built-in rate limiting and simple command interface ensure secure and controlled interaction.

Data Encryption in 6G Networks: A Zero-Knowledge Proof Model (Research Paper)

- Demonstrated effectiveness of zero-knowledge proofs (ZKPs) in enhancing network security.
- Leveraged RSA and AES for key generation to bolster data privacy in 6G network.
- Conducted simulations to validate robust communication security.

Phishing Hook

- Created a phishing detection website using deep learning techniques.
- Trained a model to analyze web pages for phishing indicators like fake login forms and misleading content.
- Developed a user-friendly interface for real-time URL analysis and detection.